INCH-POUND
MIL-PRF-19978/3F
27 May 1999
SUPERSEDING
MIL-C-19978/3E
15 April 1977

PERFORMANCE SPECIFICATION SHEET

# CAPACITORS, FIXED, PLASTIC (OR PAPER-PLASTIC) DIELECTRIC, (HERMETICALLY SEALED IN METAL CASES), STYLE CQ72

This specification is approved for use by all Departments and Agencies of Department of Defense.

The complete requirements for procuring the product described herein shall consist of this specification and MIL-PRF-19978.

Characteristic K is inactive for new design after 22 July 1991.

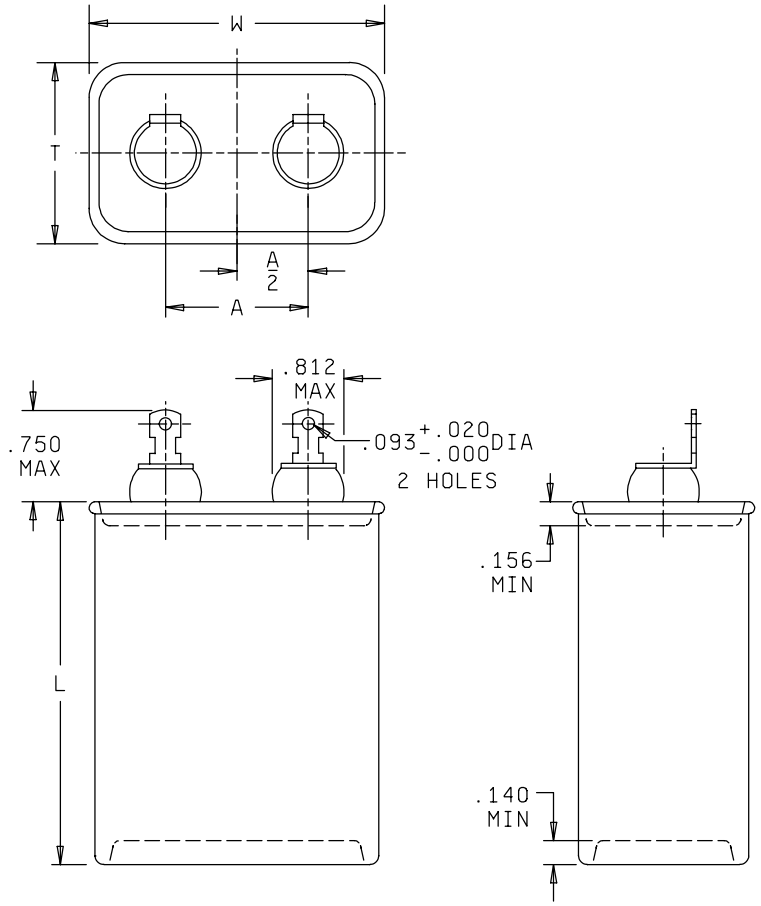



FIGURE 1. Style CQ72 capacitors.


DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

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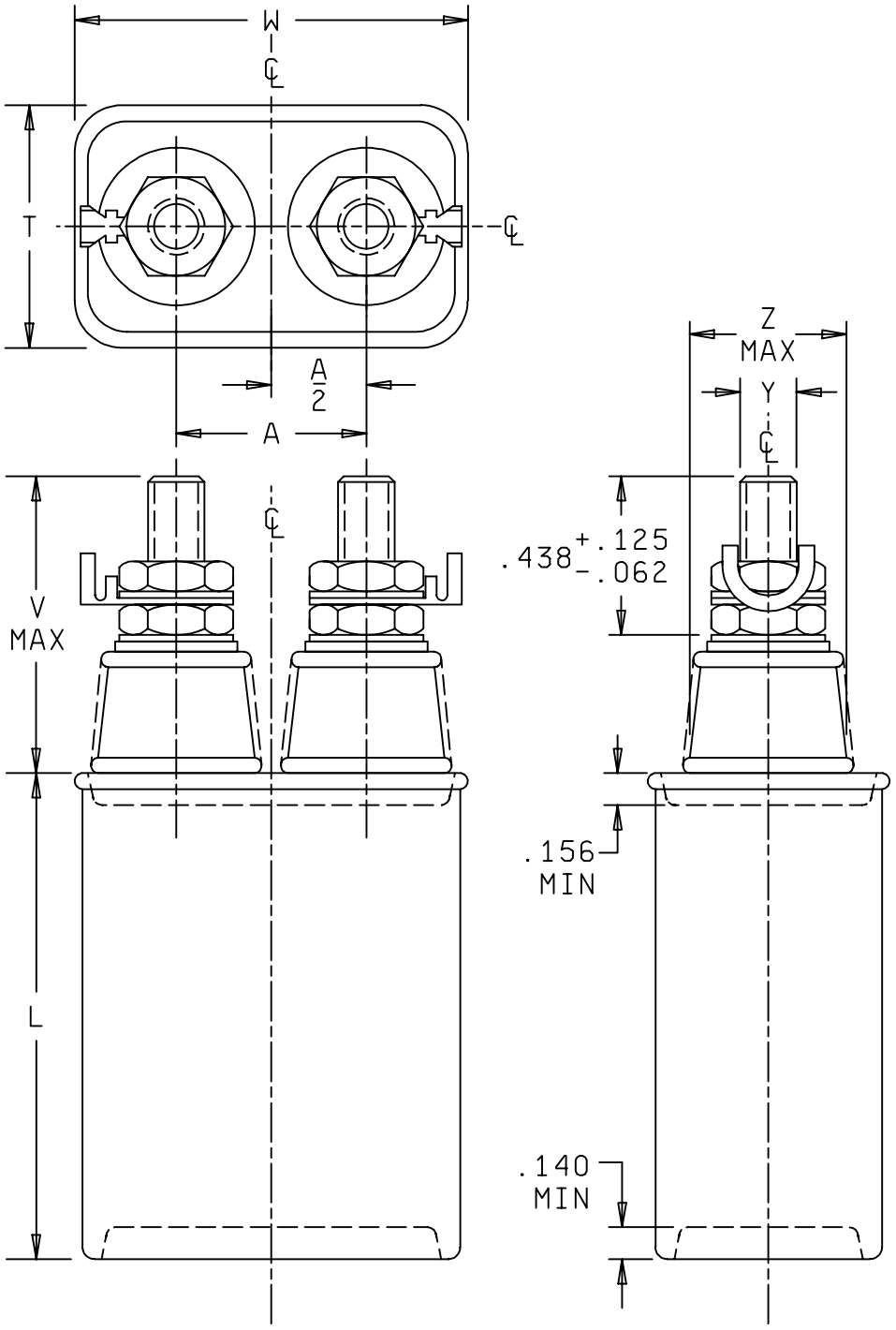



FIGURE 1. Style CQ72 capacitors - Continued.



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| Symbol | Type of terminal |
|---|---|
| B | Solder lug (nonremovable) |
| D,H | Pillar insulator (for use at altitudes up to 7,500 feet) |
| E | Pillar insulator (for use at altitudes up to 50,000 feet) |

| Terminal number | Terminal symbols | Terminal dimensions | | | |
|---|---|---|---|---|---|
| | | V, max | Y, min | Y, max | Z |
| 1 | B | .75 | | | .812 |
| 2 | E | 1.375 | .190-32 UNF-2A | .216-24 UNF-2A | .812 |
| 3 | E | 1.75 | .190-32 UNF-2A | .216-24 UNF-2A | 1.00 |
| 4 | D,E | 2.75 | .190-32 UNF-2A | .3125-18 UNF-2A | 1.438 |
| 5 | H | 4.250 | .375-16 UNF-2A | .500-13 UNF-2A | 3.000 |

| Inches | mm | Inches | mm | Inches | mm |
|---|---|---|---|---|---|
| .020 | .51 | .156 | 3.69 | 1.438 | 36.53 |
| .062 | 1.57 | .438 | 11.13 | 1.75 | 44.45 |
| .063 | 1.60 | .75 | 19.05 | 2.75 | 69.85 |
| .093 | 2.36 | .812 | 20.62 | 3.000 | 76.20 |
| .125 | 3.18 | 1.00 | 25.40 | 4.250 | 107.95 |
| .140 | 3.56 | 1.375 | 34.93 | | |

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Unless otherwise specified, tolerances are ±.005 (.13 mm) for three place decimals and ±.02 (.51 mm) for two place decimals.
4. Shape of terminal B (drilled, punches, forked, or slotted) is optional..
5. Capacitors having terminal D or E shall be supplied with necessary terminal hardware for each terminal (two nuts (one of these may be fixed cap nut on the bushing) and one lockwasher); in addition, capacitors rated at 3,000 volts dc or less shall be supplied with a solder lug capable of accommodating a wire of .063 (1.60 mm) (No. 14 AWG) for each terminal.

FIGURE 1. Style CQ720 capacitors - Continued.

REQUIREMENTS:

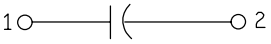
Qualification inspection for characteristic K is not required. To furnish parts to characteristic K, perform groups A and B inspections. A copy of the test data shall be forwarded to the qualifying activity.

Design and construction:

Dimensions and configuration - See figure 1 and table II.

Construction - Extended foil or laid-in-tab.

Terminals - See figure I.

CIRCUIT DIAGRAM - 


Rated voltage - See table II.



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Rated temperature: -65°C to +85°C (E)

-65°C to +125°C (K)

-55°C to +85°C (F and G)

Capacitance (cap.) (nom) - See table II.

Capacitance tolerance - ±10 percent.

Vibration grade - See table II.

Dissipation factor (DF) (max) - 1.0 percent.

Seal: Method 112 of MIL-STD-202, test condition letter A, or in accordance with MIL-PRF-19978 (alternative test).

Barometric pressure: In accordance with MIL-PRF-19978; terminal D capacitors not applicable.

Test condition letter B (3.44 inches of mercury - 50,000 feet).

Test points - Between ungrounded terminal and case.

Test potential - 125 percent of dc rated voltage.

Insulation resistance (IR):

Terminal to terminal - See table I.

Terminals to case - Greater than 10,000 megohms.

Vibration:

Low frequency (grade 1) - Method 201 of MIL-STD-202.

High frequency (grade 3) - Method 204 of MIL-STD-202, test condition B, with the following exception:
Direction and duration of motion - 4 hours in each of two mutually perpendicular directions (total of 8 hours), one parallel and the other perpendicular to the cylindrical axis.

Salt spray: In accordance with MIL-PRF-19978.

Solderability: Not applicable.

Terminal strength: Method 211 of MIL-STD-202, test condition letters A and E.

Stability at low and high temperatures:

Low temperature:

Test temperature - -55°C (F and G) or -65°C (E and K).

Capacitance change (max) - -10 percent.

High temperature:

Test temperature - +85°C (E, F, and G) or +125°C (K).

Capacitance change (max) - +10 percent.



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Life:

Capacitance change (max) - +10 percent.

Resistance to soldering heat: Not applicable.

TABLE I. Terminal-to-terminal insulation resistance.

| Capacitance rating | Minimum insulation resistance |
|---|---|
| Characteristic E | |
| | At 25°C |
| 0 to 0.6 microfarad | 25,000 megohms |
| Greater than 0.6 microfarad | 15,000 megohm-microfarads 1/ |
| | At 85°C |
| 0 to 0.15 microfarad | 2,000 megohms |
| Greater than 0.15 microfarad | 300 megohm-microfarads 1/ |
| Characteristic K | |
| | At 25°C |
| 0 to 0.6 microfarad | 25,000 megohms |
| Greater than 0.6 microfarad | 15,000 megohm-microfarads 1/ |
| | At 125°C |
| 0 to 0.08 microfarad | 250 megohms |
| Greater than 0.08 microfarad | 20 megohm-microfarads 1/ |
| Characteristic F | |
| | At 25°C |
| 0.33 microfarad and less | 6,000 megohms |
| Greater than 0.33 microfarad | 2,000 megohm-microfarads 1/ |
| | At 85°C |
| 0.033 microfarad and less | 600 megohms |
| Greater than 0.033 microfarad | 20 megohm-microfarads 1/ |
| Characteristic G | |
| | At 25°C |
| 0.33 microfarad and less | 4,500 megohms |
| Greater than 0.33 microfarad | 1,500 megohm-microfarads 1/ |
| | At 85°C |
| 0.033 microfarad and less | 450 megohms |
| Greater than 0.033 microfarad | 15 megohm-microfarads 1/ |

1/ Product obtained by multiplying the capacitance in microfarads by the insulation resistance in megohms.

TABLE II. Style CQ72 capacitors.

| Part number 1/ | DC voltage rating (Volts) | Capacitance (μF) | Vibration grade | Case size | | | | Available terminal identification number | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | Characteristic | | | | Terminal | | | |
| | | | | E | K | F | G | B | D | E | H |
| CQ72-1KE254K3 | 400 | 0.25 | 3 | -- | A1 | -- | -- | 1 | --- | 2 | --- |
| CQ72-1-E504K3 | 400 | 0.5 | 3 | A1 | A2 | -- | -- | 1 | --- | 2 | --- |
| CQ72-1-E105K3 | 400 | 1.0 | 3 | A1 | A3 | -- | -- | 1 | --- | 2 | --- |
| CQ72-1-E205K3 | 400 | 2 | 3 | A2 | A5 | -- | -- | 1 | --- | 2 | --- |
| CQ72-1-E405K3 | 400 | 4 | 3 | B2 | B6 | -- | -- | 1 | --- | 2 | --- |
| CQ72-1-E605K3 | 400 | 6 | 3 | B6 | C3 | -- | -- | 1 | --- | 2 | --- |
| CQ72-1-E805K3 | 400 | 8 | 3 | C3 | D4 | -- | -- | 1 | --- | 2 | --- |
| CQ72-1-E106K3 | 400 | 10 | 3 | C4 | E4 | -- | -- | 1 | --- | 2 | --- |

See footnote at end of table.



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TABLE II. Style CQ72 capacitors - Continued.

| Part number 1/ | DC voltage rating (Volts) | Capacitance (µF) | Vibration grade | Case size | | | | Available terminal identification number | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | Characteristic | | | | Terminal | | | |
| | | | | E | K | F | G | B | D | E | H |
| CQ72-1-E126K3 | 400 | 12 | 3 | D2 | E6 | -- | -- | 1 | --- | 2 | --- |
| CQ72-1-E156K3 | 400 | 15 | 3 | D5 | G4 | -- | -- | 1 | --- | 2 | --- |
| CQ72-1FF254K1 | 600 | 0.25 | 1 | -- | -- | A1 | -- | 1 | --- | 2 | --- |
| CQ72-1-F504K3 | 600 | 0.5 | 3 | A1 | A2 | -- | -- | 1 | --- | 2 | --- |
| CQ72-1-F105K- | 600 | 1.0 | 1,3 | A2 | A3 | -- | -- | 1 | --- | 2 | --- |
| CQ72-1-F205K- | 600 | 2 | 1,3 | A2 | B4 | A4 | A3 | 1 | --- | 2 | --- |
| CQ72-1-F405K- | 600 | 4 | 1,3 | B3 | D2 | B4 | B3 | 1 | --- | 2 | --- |
| CQ72-1-F605K- | 600 | 6 | 1,3 | B6 | E4 | C2 | B6 | 1 | --- | 2 | --- |
| CQ72-1-F805K- | 600 | 8 | 1,3 | C2 | F1 | C3 | C2 | 1 | --- | 2 | --- |
| CQ72-1-F106K- | 600 | 10 | 1,3 | C4 | J2 | D3 | C4 | 1 | --- | 2 | --- |
| CQ72-1-F126K3 | 600 | 12 | 3 | E2 | J3 | -- | -- | 1 | --- | 2 | --- |
| CQ72-1-F156K3 | 600 | 15 | 3 | E5 | J6 | -- | -- | 1 | --- | 2 | --- |
| CQ72-1FG104K1 | 1,000 | 0.1 | 1 | -- | -- | A1 | -- | 1 | --- | 2 | --- |
| CQ72-1-G254K- | 1,000 | 0.25 | 1,3 | -- | A2 | A2 | -- | 1 | --- | 2 | --- |
| CQ72-1-G504K- | 1,000 | 0.5 | 1,3 | A1 | A3 | A2 | -- | 1 | --- | 2 | --- |
| CQ72-1-G105K- | 1,000 | 1.0 | 1,3 | A2 | B3 | A3 | A2 | 1 | --- | 2 | --- |
| CQ72-1-G205K- | 1,000 | 2 | 1,3 | A5 | C4 | A6 | A5 | 1 | --- | 2 | --- |
| CQ72-1-G405K- | 1,000 | 4 | 1,3 | B6 | F1 | C2 | B6 | 1 | --- | 2 | --- |
| CQ72-1-G605K- | 1,000 | 6 | 1,3 | C4 | G2 | D3 | C4 | 1 | --- | 2 | --- |
| CQ72-1-G805K- | 1,000 | 8 | 1,3 | D2 | J2 | D5 | D2 | 1 | --- | 2 | --- |
| CQ72-1-G106K- | 1,000 | 10 | 1,3 | D5 | J5 | E5 | D5 | 1 | --- | 2 | --- |
| CQ72-1-G126K- | 1,000 | 12 | 1,3 | E4 | J6 | E6 | E4 | 1 | --- | 2 | --- |
| CQ72-1-G156K- | 1,000 | 15 | 1,3 | F1 | J11 | G2 | F1 | 1 | --- | 2 | --- |
| CQ72-1-H104K- | 1,500 | 0.1 | 1,3 | A1 | A1 | A2 | -- | 1 | --- | 2 | --- |
| CQ72-1-H254K- | 1,500 | 0.25 | 1,3 | A1 | A2 | A3 | A2 | 1 | --- | 2 | --- |
| CQ72-1-H504K- | 1,500 | 0.5 | 1,3 | A2 | A5 | A3 | -- | 1 | --- | 2 | --- |
| CQ72-1-H105K- | 1,500 | 1.0 | 1,3 | A5 | B5 | A6 | A5 | 1 | --- | 2 | --- |
| CQ72-1-H205K- | 1,500 | 2 | 1,3 | B6 | D4 | B6 | B5 | 1 | --- | 2 | --- |
| CQ72-1-H405K- | 1,500 | 4 | 1,3 | D5 | G4 | D4 | C4 | 1 | --- | 2 | --- |
| CQ72-1-H605K- | 1,500 | 6 | 1,3 | G1 | J6 | E5 | D5 | 1 | --- | 2 | --- |
| CQ72-1-H805K- | 1,500 | 8 | 1,3 | J2 | J10 | F2 | F1 | 1 | --- | 2 | --- |
| CQ72-1-H106K- | 1,500 | 10 | 1,3 | J4 | -- | G4 | G2 | 1 | --- | 2 | --- |
| CQ72-1-H126K- | 1,500 | 12 | 1,3 | J6 | -- | J3 | G2 | 1 | --- | 2 | --- |
| CQ72-1-H156K- | 1,500 | 15 | 1,3 | J10 | -- | J6 | J3 | 1 | --- | 2 | --- |
| CQ72-1FJ254K1 | 2,000 | 0.25 | 1 | -- | -- | B1 | -- | --- | --- | 3 | --- |
| CQ72-1FJ504K1 | 2,000 | 0.5 | 1 | -- | -- | B2 | -- | --- | --- | 3 | --- |
| CQ72-1-J105K- | 2,000 | 1.0 | 1,3 | -- | C3 | B6 | B3 | --- | --- | 3 | --- |
| CQ72-1-J205K- | 2,000 | 2 | 1,3 | C4 | E6 | C4 | C3 | --- | --- | 3 | --- |
| CQ72-1-J405K- | 2,000 | 4 | 1,3 | F1 | J4 | E6 | E3 | --- | --- | 3 | --- |
| CQ72-1-J605K- | 2,000 | 6 | 1,3 | G3 | J8 | G4 | G1 | --- | --- | 3 | --- |
| CQ72-1-J805K- | 2,000 | 8 | 1,3 | J2 | -- | J4 | G3 | --- | --- | 3 | --- |
| CQ72-1-J106K- | 2,000 | 10 | 1,3 | J5 | -- | J7 | J3 | --- | --- | 3 | --- |
| CQ72-1-J126K- | 2,000 | 12 | 1,3 | J6 | -- | J9 | J5 | --- | --- | 3 | --- |
| CQ72-1-J156K- | 2,000 | 15 | 1,3 | J11 | -- | J10 | J7 | --- | --- | 3 | --- |
| CQ72-1FK104K1 | 2,500 | 0.1 | 1 | -- | -- | B1 | -- | --- | --- | 3 | --- |
| CQ72-1-K254K1 | 2,500 | 0.25 | 1 | -- | -- | B2 | B1 | --- | --- | 3 | --- |
| CQ72-1-K504K- | 2,500 | 0.5 | 1,3 | -- | B5 | B4 | B3 | --- | --- | 3 | --- |
| CQ72-1-K105K- | 2,500 | 1.0 | 1,3 | -- | D2 | D2 | D1 | --- | --- | 3 | --- |
| CQ72-1-K205K- | 2,500 | 2 | 1,3 | C4 | G4 | E5 | D5 | --- | --- | 3 | --- |
| CQ72-1-K405K- | 2,500 | 4 | 1,3 | E6 | J8 | J2 | -- | --- | --- | 3 | --- |

See footnote at end of table.



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TABLE II. Style CQ72 capacitors - Continued.

| Part number 1/ | DC voltage rating (Volts) | Capacitance (µF) | Vibration grade | Case size | | | | Available terminal identification number | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | Characteristic | | | | Terminal | | | |
| | | | | E | K | F | G | B | D | E | H |
| CQ72-1-K605K- | 2,500 | 6 | 1,3 | J2 | -- | J6 | J4 | --- | --- | 3 | --- |
| CQ72-1-K805K- | 2,500 | 8 | 1,3 | J5 | -- | J8 | J6 | --- | --- | 3 | --- |
| CQ72-1-K106K- | 2,500 | 10 | 1,3 | J7 | -- | J10 | J8 | --- | --- | 3 | --- |
| CQ72-1-K126K- | 2,500 | 12 | 1,3 | J11 | -- | K3 | J10 | --- | --- | 3 | --- |
| CQ72-L104K1 | 3,000 | 0.1 | 1 | --- | --- | B2 | B1 | --- | --- | 3 | --- |
| CQ72-L254K1 | 3,000 | 0.25 | 1 | --- | --- | B3 | B2 | --- | --- | 3 | --- |
| CQ72-L504K1 | 3,000 | 0.5 | 1 | --- | --- | B6 | B5 | --- | --- | 3 | --- |
| CQ72-L105K- | 3,000 | 1.0 | 1,3 | D1 | E4 | E4 | E3 | --- | --- | 3 | --- |
| CQ72-L205K- | 3,000 | 2 | 1,3 | E3 | J2 | G2 | G1 | --- | --- | 3 | --- |
| CQ72-L405K- | 3,000 | 4 | 1,3 | G4 | J11 | J7 | J3 | --- | --- | 3 | --- |
| CQ72-L605K- | 3,000 | 6 | 1,3 | J5 | --- | K2 | J9 | --- | --- | 3 | --- |
| CQ72E1FM104K1 | 4,000 | 0.1 | 1 | -- | -- | E1 | -- | --- | --- | 4 | --- |
| CQ72E1-M254K- | 4,000 | 0.25 | 1,3 | -- | C1 | E3 | E1 | --- | --- | 4 | --- |
| CQ72E1-M504K- | 4,000 | 0.5 | 1,3 | -- | C4 | E5 | E3 | --- | --- | 4 | --- |
| CQ72E1-M105K- | 4,000 | 1.0 | 1,3 | C2 | F2 | G2 | E6 | --- | --- | 4 | --- |
| CQ72E1-M205K- | 4,000 | 2 | 1,3 | E6 | J7 | J6 | J4 | --- | --- | 4 | --- |
| CQ72E1-M405K- | 4,000 | 4 | 1,3 | J5 | -- | J11 | J10 | --- | --- | 4 | --- |
| CQ72E1-N104K- | 5,000 | 0.1 | 1,3 | C1 | -- | E1 | -- | --- | --- | 4 | --- |
| CQ72E1-N254K- | 5,000 | 0.25 | 1,3 | C1 | -- | E3 | -- | --- | --- | 4 | --- |
| CQ72E1-N504K- | 5,000 | 0.5 | 1,3 | C1 | E4 | E6 | E4 | --- | --- | 4 | --- |
| CQ72E1-N105K- | 5,000 | 1.0 | 1,3 | D2 | G4 | J4 | J2 | --- | --- | 4 | --- |
| CQ72E1-N205K- | 5,000 | 2 | 1,3 | F2 | J11 | J11 | J6 | --- | --- | 4 | --- |
| CQ72E1-N405K1 | 5,000 | 4 | 1 | -- | -- | K4 | K3 | --- | --- | 4 | --- |
| CQ72D1-P104K- | 6,000 | 0.1 | 1,3 | C1 | -- | E3 | E2 | --- | 4 | --- | --- |
| CQ72D1-P254K- | 6,000 | 0.25 | 1,3 | C1 | D2 | E5 | E4 | --- | 4 | --- | --- |
| CQ72D1-P504K- | 6,000 | 0.5 | 1,3 | C1 | E6 | J3 | J2 | --- | 4 | --- | --- |
| CQ72D1-P105K- | 6,000 | 1.0 | 1,3 | E5 | J6 | J11 | J9 | --- | 4 | --- | --- |
| CQ72D1-P205K- | 6,000 | 2 | 1,3 | G1 | -- | L1 | K3 | --- | 4 | --- | --- |
| CQ72D1-R104K- | 7,500 | 0.1 | 1,3 | C1 | C1 | E5 | E3 | --- | 4 | --- | --- |
| CQ72D1-R254K- | 7,500 | 0.25 | 1,3 | C2 | G1 | J2 | J1 | --- | 4 | --- | --- |
| CQ72D1-R504K- | 7,500 | 0.5 | 1,3 | E4 | J3 | J6 | J4 | --- | 4 | --- | --- |
| CQ72D1-R105K- | 7,500 | 1.0 | 1,3 | G3 | -- | K3 | -- | --- | 4 | --- | --- |
| CQ72D1-R205K- | 7,500 | 2 | 1,3 | J10 | -- | L2 | K4 | --- | 4 | --- | --- |
| CQ72D1-S104K- | 10,000 | 0.1 | 1,3 | C2 | E2 | -- | -- | --- | 4 | --- | --- |
| CQ72D1-S254K- | 10,000 | 0.25 | 1,3 | E3 | J3 | -- | -- | --- | 4 | --- | --- |
| CQ72-1-S504K- | 10,000 | 0.5 | 1,3 | F2 | J9 | K4 | K3 | --- | 4 | --- | 5 |
| CQ72-1-S105K- | 10,000 | 1.0 | 1,3 | J6 | -- | L2 | K4 | --- | 4 | --- | 5 |
| CQ72-1-S205K1 | 10,000 | 2 | 1 | -- | -- | M1 | L3 | --- | 4 | --- | 5 |
| CQ72-1-T104K- | 12,500 | 0.1 | 1,3 | E2 | J1 | K1 | -- | --- | 4 | --- | 5 |
| CQ72-1-T254K- | 12,500 | 0.25 | 1,3 | E4 | J5 | K2 | -- | --- | 4 | --- | 5 |
| CQ72-1-T504K- | 12,500 | 0.5 | 1,3 | J4 | -- | L1 | K4 | --- | 4 | --- | 5 |
| CQ72-1-T105K- | 12,500 | 1.0 | 1,3 | J10 | -- | M1 | L3 | --- | 4 | --- | 5 |
| CQ72-1-T205K1 | 12,500 | 2 | 1 | -- | -- | M2 | M1 | --- | 4 | --- | 5 |

1/ Complete part number shall include additional symbols to indicate terminal, characteristic, and vibration grade, as applicable.



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TABLE III. Case dimensions and retainer part number.

| Case size 1/ | Case dimensions (in inches) | | | | Retainer part number 2/ |
|---|---|---|---|---|---|
| | W ±.062 | T ±.062 | L ±.062 | A +.0625, -.187 | |
| A1 | 1.812 | 1.062 | 1.625 | 0.812 | CQ072-A1 |
| A2 | 1.812 | 1.062 | 2.250 | 0.812 | CQ072-A2 |
| A3 | 1.812 | 1.062 | 2.875 | 0.812 | CQ072-A3 |
| A4 | 1.812 | 1.062 | 3.250 | 0.812 | CQ072-A4 |
| A5 | 1.812 | 1.062 | 3.875 | 0.812 | CQ072-A5 |
| A6 | 1.812 | 1.062 | 4.750 | 0.812 | CQ072-A6 |
| B1 | 2.500 | 1.188 | 2.500 | 1.125 | CQ072-B1 |
| B2 | 2.500 | 1.188 | 2.875 | 1.125 | CQ072-B2 |
| B3 | 2.500 | 1.188 | 3.500 | 1.125 | CQ072-B3 |
| B4 | 2.500 | 1.188 | 3.875 | 1.125 | CQ072-B4 |
| B5 | 2.500 | 1.188 | 4.250 | 1.125 | CQ072-B5 |
| B6 | 2.500 | 1.188 | 4.250 | 1.125 | CQ072-B6 |
| C1 | 3.750 | 1.250 | 3.250 | 2.000 | CQ072-C1 |
| C2 | 3.750 | 1.250 | 3.875 | 2.000 | CQ072-C2 |
| C3 | 3.750 | 1.250 | 4.250 | 2.000 | CQ072-C3 |
| C4 | 3.750 | 1.250 | 4.750 | 2.000 | CQ072-C4 |
| D1 | 3.750 | 1.750 | 3.250 | 2.000 | CQ072-D1 |
| D2 | 3.750 | 1.750 | 3.875 | 2.000 | CQ072-D2 |
| D3 | 3.750 | 1.750 | 4.000 | 2.000 | CQ072-D3 |
| D4 | 3.750 | 1.750 | 4.250 | 2.000 | CQ072-D4 |
| D5 | 3.750 | 1.750 | 4.750 | 2.000 | CQ072-D5 |
| E1 | 3.750 | 2.250 | 2.750 | 2.000 | CQ072-E1 |
| E2 | 3.750 | 2.250 | 3.375 | 2.000 | CQ072-E2 |
| E3 | 3.750 | 2.250 | 3.875 | 2.000 | CQ072-E3 |
| E4 | 3.750 | 2.250 | 4.500 | 2.000 | CQ072-E4 |
| E5 | 3.750 | 2.250 | 4.750 | 2.000 | CQ072-E5 |
| E6 | 3.750 | 2.250 | 5.125 | 2.000 | CQ072-E6 |
| F1 | 3.750 | 2.500 | 4.750 | 2.000 | CQ072-F1 |
| F2 | 3.750 | 2.500 | 5.750 | 2.000 | CQ072-F2 |
| G1 | 3.750 | 3.188 | 4.500 | 2.000 | CQ072-G1 |
| G2 | 3.750 | 3.188 | 4.750 | 2.000 | CQ072-G2 |
| G3 | 3.750 | 3.188 | 5.125 | 2.000 | CQ072-G3 |
| G4 | 3.750 | 3.188 | 5.500 | 2.000 | CQ072-G4 |
| J1 | 4.562 | 3.750 | 3.875 | 2.000 | CQ072-J1 |
| J2 | 4.562 | 3.750 | 4.375 | 2.000 | CQ072-J2 |
| J3 | 4.562 | 3.750 | 4.750 | 2.000 | CQ072-J3 |
| J4 | 4.562 | 3.750 | 5.125 | 2.000 | CQ072-J4 |
| J5 | 4.562 | 3.750 | 5.500 | 2.000 | CQ072-J5 |
| J6 | 4.562 | 3.750 | 6.000 | 2.000 | CQ072-J6 |
| J7 | 4.562 | 3.750 | 6.500 | 2.000 | CQ072-J7 |
| J8 | 4.562 | 3.750 | 7.000 | 2.000 | CQ072-J8 |
| J9 | 4.562 | 3.750 | 7.500 | 2.000 | CQ072-J9 |
| J10 | 4.562 | 3.750 | 8.500 | 2.000 | CQ072-J10 |
| J11 | 4.562 | 3.750 | 9.625 | 2.000 | CQ072-J11 |
| K1 | 8.000 | 4.000 | 5.500 | 4.500 | CQ072-K1 |
| K2 | 8.000 | 4.000 | 7.000 | 4.500 | CQ072-K2 |
| K3 | 8.000 | 4.000 | 9.250 | 4.500 | CQ072-K3 |
| K4 | 8.000 | 4.000 | 11.000 | 4.500 | CQ072-K4 |
| L1 | 13.500 | 4.125 | 7.000 | 6.750 | CQ072-L1 |
| L2 | 13.500 | 4.125 | 9.250 | 6.750 | CQ072-L2 |
| L3 | 13.500 | 4.125 | 13.125 | 6.750 | CQ072-L3 |

See footnotes at end of table.



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TABLE III. Case dimensions and retainer part number - Continued.

| Case size 1/ | Case dimensions (in inches) | | | | Retainer part number 2/ |
|---|---|---|---|---|---|
| | W ±.062 | T ±.062 | L ±.062 | A +.0625, -.187 | |
| M1 | 13.500 | 5.125 | 12.875 | 6.750 | CQ072-M1 |
| M2 | 13.500 | 5.125 | 13.750 | 6.750 | CQ072-M2 |

1/ For case size J, the retainers are attached to the sides of the case which are parallel to the centerline of the terminals.
2/ Complete retainer part number will include the symbol "F" for footed or "S" for the spaded design.

TABLE IV. CQ072 retainer dimensions.

| Retainer part number 1/ | D ±.031 | | E max | G max | H ±.016 2/ | J | L | | M ±.016 | T ±.062 | Case size |
|---|---|---|---|---|---|---|---|---|---|---|---|
| | Footed retainer | Spade-lug retainer | | | | | Footed retainer | Spade-lug retainer | | | |
| CQ072-A1 | 2.250 | 2.062 | 2.719 | .625 | N/A | .187 | 1.125 | 1.375 | .391 | 2.375 | A1 |
| CQ072-A2 | 2.250 | 2.062 | 2.719 | .625 | N/A | .187 | 2.125 | 2.000 | .391 | 3.000 | A2 |
| CQ072-A3 | 2.250 | 2.062 | 2.719 | .625 | N/A | .187 | 2.750 | 2.625 | .391 | 3.625 | A3 |
| CQ072-A4 | 2.250 | 2.062 | 3.062 | .906 | N/A | .187 | 3.125 | 3.000 | .391 | 4.000 | A4 |
| CQ072-A5 | 2.250 | 2.062 | 2.719 | .625 | N/A | .187 | 3.750 | 3.625 | .391 | 4.625 | A5 |
| CQ072-A6 | 2.250 | 2.062 | 2.719 | .625 | N/A | .187 | 4.625 | 4.500 | .391 | 5.500 | A6 |
| CQ072-B1 | 3.000 | 2.750 | 3.812 | .906 | N/A | .187 | 2.375 | 2.250 | .391 | 3.250 | B1 |
| CQ072-B2 | 3.000 | 2.750 | 3.938 | .625 | N/A | .187 | 2.750 | 2.625 | .391 | 3.625 | B2 |
| CQ072-B3 | 3.000 | 2.750 | 3.938 | .625 | N/A | .187 | 3.375 | 3.250 | .391 | 4.250 | B3 |
| CQ072-B4 | 3.000 | 2.750 | 3.938 | .625 | N/A | .187 | 3.750 | 3.625 | .391 | 4.625 | B4 |
| CQ072-B5 | 3.000 | 2.750 | 3.938 | .625 | N/A | .187 | 4.125 | 4.000 | .391 | 5.000 | B5 |
| CQ072-B6 | 3.000 | 2.750 | 3.938 | .625 | N/A | .187 | 4.625 | 4.500 | .391 | 5.500 | B6 |
| CQ072-C1 | 4.375 | 4.000 | 4.844 | .625 | N/A | .187 | 3.125 | 3.000 | .391 | 4.000 | C1 |
| CQ072-C2 | 4.375 | 4.000 | 4.844 | .625 | N/A | .187 | 3.750 | 3.625 | .391 | 4.625 | C2 |
| CQ072-C3 | 4.375 | 4.000 | 4.844 | .625 | N/A | .187 | 4.125 | 4.000 | .391 | 5.000 | C3 |
| CQ072-C4 | 4.375 | 4.000 | 4.844 | .625 | N/A | .187 | 4.625 | 4.500 | .391 | 5.500 | C4 |
| CQ072-D1 | 4.375 | 4.000 | 4.969 | 1.375 | .625 | .213 | 3.125 | 3.000 | .844 | 4.000 | D1 |
| CQ072-D2 | 4.375 | 4.000 | 4.969 | 1.375 | .625 | .213 | 3.750 | 3.625 | .844 | 4.625 | D2 |
| CQ072-D3 | 4.375 | 4.000 | 5.000 | 1.469 | .625 | .213 | 3.875 | 3.750 | .844 | 4.750 | D3 |
| CQ072-D4 | 4.375 | 4.000 | 4.969 | 1.375 | .625 | .213 | 4.125 | 4.000 | .844 | 5.000 | D4 |
| CQ072-D5 | 4.375 | 4.000 | 4.969 | 1.375 | .625 | .213 | 4.625 | 4.500 | .844 | 5.500 | D5 |
| CQ072-E1 | 4.375 | 4.000 | 5.250 | 1.967 | 1.250 | .213 | 2.625 | 2.500 | 1.344 | 3.500 | E1 |
| CQ072-E2 | 4.375 | 4.000 | 4.969 | 1.875 | 1.250 | .213 | 3.250 | 3.125 | 1.344 | 4.125 | E2 |
| CQ072-E3 | 4.375 | 4.000 | 4.969 | 1.875 | 1.250 | .213 | 3.750 | 3.625 | 1.344 | 4.625 | E3 |
| CQ072-E4 | 4.375 | 4.000 | 4.969 | 1.875 | 1.250 | .213 | 4.375 | 4.250 | 1.344 | 5.250 | E4 |
| CQ072-E5 | 4.375 | 4.000 | 4.969 | 1.875 | 1.250 | .213 | 4.625 | 4.500 | 1.344 | 5.500 | E5 |
| CQ072-E6 | 4.375 | 4.000 | 4.969 | 1.875 | 1.250 | .213 | 5.000 | 4.875 | 1.344 | 5.875 | E6 |
| CQ072-F1 | 4.375 | 4.000 | 4.969 | 1.875 | 1.250 | .213 | 4.625 | 4.500 | 1.344 | 5.500 | F1 |
| CQ072-F2 | 4.375 | 4.000 | 4.969 | 1.875 | 1.250 | .213 | 5.625 | 5.500 | 1.344 | 6.500 | F2 |
| CQ072-G1 | 4.375 | 4.000 | 4.969 | 2.812 | 1.938 | .213 | 4.375 | 4.250 | 2.281 | 5.250 | G1 |
| CQ072-G2 | 4.375 | 4.000 | 4.969 | 2.812 | 1.938 | .213 | 4.625 | 4.500 | 2.281 | 5.500 | G2 |
| CQ072-G3 | 4.375 | 4.000 | 4.969 | 2.812 | 1.938 | .213 | 5.000 | 4.875 | 2.281 | 5.875 | G3 |
| CQ072-G4 | 4.375 | 4.000 | 4.969 | 2.812 | 1.938 | .213 | 5.375 | 5.250 | 2.281 | 6.250 | G4 |
| CQ072-J1 | 4.375 | 4.000 | 4.969 | 4.000 | 3.375 | .281 | 3.750 | 3.625 | 3.656 | 4.625 | J1 |
| CQ072-J2 | 4.375 | 4.000 | 4.969 | 4.000 | 3.375 | .281 | 4.250 | 4.125 | 3.656 | 5.125 | J2 |
| CQ072-J3 | 4.375 | 4.000 | 4.969 | 4.000 | 3.375 | .281 | 4.625 | 4.500 | 3.656 | 5.500 | J3 |
| CQ072-J4 | 4.375 | 4.000 | 4.969 | 4.000 | 3.375 | .281 | 5.000 | 4.875 | 3.656 | 5.875 | J4 |

See footnote at end of table.



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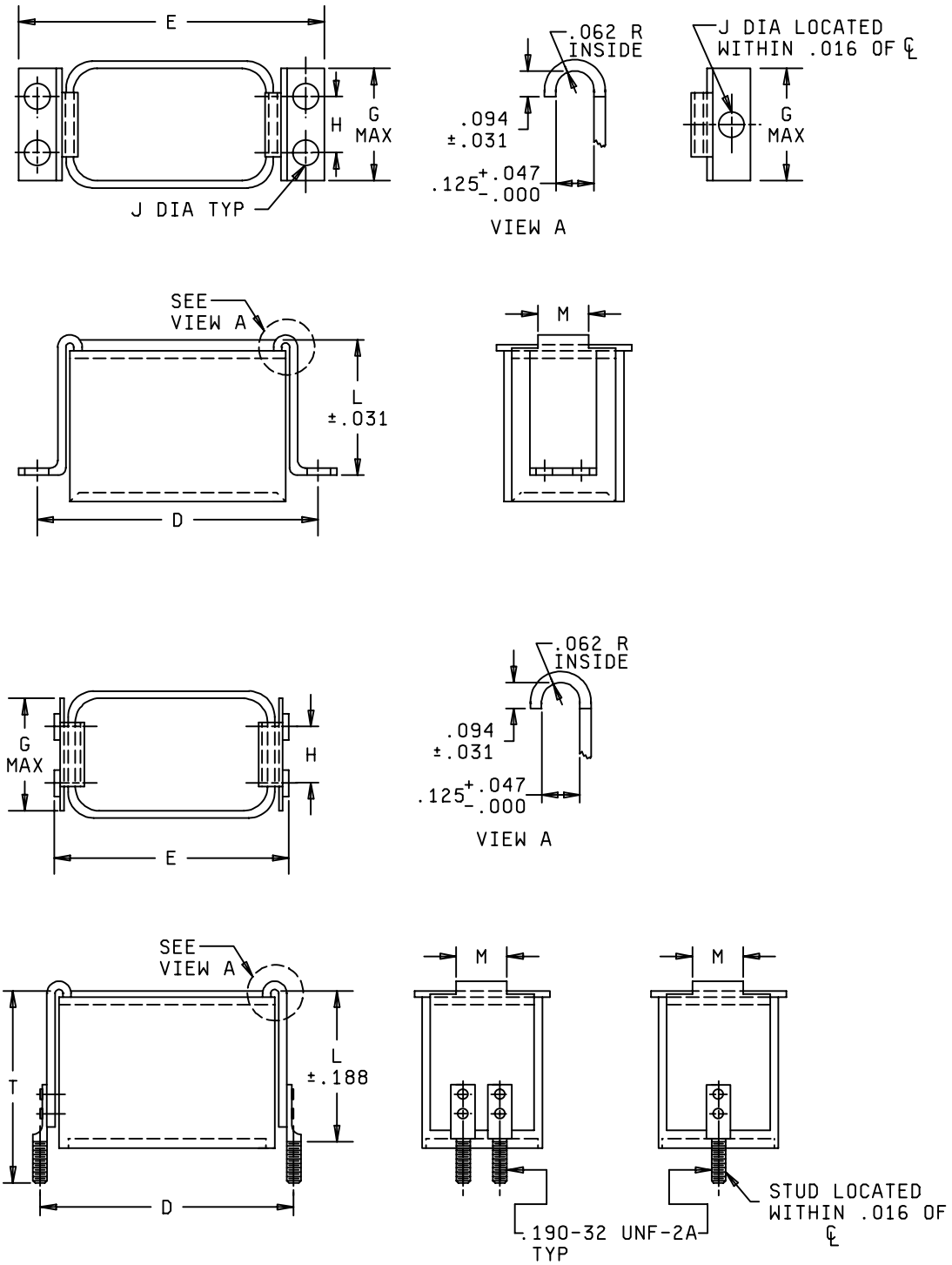
TABLE IV. CQ072 retainer dimensions - Continued.

| Retainer part number 1/ | D ±.031 | | E max | G max | H ±.016 2/ | J | L | | M ±.016 | T ±.062 | Case size |
|---|---|---|---|---|---|---|---|---|---|---|---|
| | Footed retainer | Spade-lug retainer | | | | | Footed retainer | Spade-lug retainer | | | |
| CQ072-J5 | 4.375 | 4.000 | 4.969 | 4.000 | 3.375 | .281 | 5.375 | 5.250 | 3.656 | 6.250 | J5 |
| CQ072-J6 | 4.375 | 4.000 | 4.969 | 4.000 | 3.375 | .281 | 5.875 | 5.750 | 3.656 | 6.750 | J6 |
| CQ072-J7 | 4.500 | 4.000 | 5.094 | 4.000 | 3.375 | .281 | 6.375 | 6.250 | 3.656 | 7.250 | J7 |
| CQ072-J8 | 4.500 | 4.000 | 5.094 | 4.000 | 3.375 | .281 | 6.875 | 6.750 | 3.656 | 7.750 | J8 |
| CQ072-J9 | 4.500 | 4.000 | 5.094 | 4.000 | 3.375 | .281 | 7.375 | 7.250 | 3.656 | 8.250 | J9 |
| CQ072-J10 | 4.500 | 4.000 | 5.094 | 4.000 | 3.375 | .281 | 8.375 | 8.250 | 3.656 | 9.250 | J10 |
| CQ072-J11 | 4.500 | 4.000 | 5.094 | 4.000 | 3.375 | .281 | 9.500 | 9.375 | 3.656 | 10.375 | J11 |
| CQ072-K1 | 10.125 | N/A | 11.250 | 4.375 | 2.125 | .438 | 5.250 | N/A | 3.656 | N/A | K1 |
| CQ072-K2 | 10.125 | N/A | 11.250 | 4.375 | 2.125 | .438 | 6.750 | N/A | 3.656 | N/A | K2 |
| CQ072-K3 | 10.125 | N/A | 11.250 | 4.375 | 2.125 | .438 | 9.000 | N/A | 3.656 | N/A | K3 |
| CQ072-K4 | 10.125 | N/A | 11.250 | 4.375 | 2.125 | .438 | 10.750 | N/A | 3.656 | N/A | K4 |
| CQ072-L1 | 15.625 | N/A | 16.750 | 4.500 | 2.125 | .438 | 6.750 | N/A | 3.656 | N/A | L1 |
| CQ072-L2 | 15.625 | N/A | 16.750 | 4.500 | 2.125 | .438 | 9.000 | N/A | 3.656 | N/A | L2 |
| CQ072-L3 | 15.625 | N/A | 16.750 | 4.500 | 2.125 | .438 | 12.875 | N/A | 3.656 | N/A | L3 |
| CQ072-M1 | 15.625 | N/A | 16.750 | 5.500 | 3.125 | .438 | 12.625 | N/A | 3.656 | N/A | M1 |
| CQ072-M2 | 15.625 | N/A | 16.750 | 5.500 | 3.125 | .438 | 13.500 | N/A | 3.656 | N/A | M2 |

1/ Complete retainer part number shall include additional symbol to indicate design of retainer ("S" spaded, "F" footed).
2/ Single hole or spade bolt located within .016 of center of retainer.



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CQ072 SPADE -LUG RETAINERS
(For use with style CQ72 Case sizes A1 through M2)

FIGURE 2. CQ072 retainers.



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TABLE V. Millimeter equivalent of decimal inches.

| Inches | mm | Inches | mm | Inches | mm | Inches | mm |
|---|---|---|---|---|---|---|---|
| .015 | 0.38 | 1.625 | 41.28 | 3.875 | 98.43 | 7.000 | 177.80 |
| .016 | 0.41 | 1.750 | 44.45 | 3.938 | 100.03 | 7.250 | 184.15 |
| .025 | 0.64 | 1.812 | 46.02 | 4.000 | 101.60 | 7.375 | 187.33 |
| .031 | 0.79 | 1.938 | 49.23 | 4.125 | 104.78 | 7.500 | 190.50 |
| .045 | 1.14 | 1.967 | 49.96 | 4.250 | 107.95 | 7.750 | 196.85 |
| .047 | 1.19 | 2.000 | 50.80 | 4.375 | 111.13 | 8.000 | 203.20 |
| .062 | 1.57 | 2.062 | 52.37 | 4.500 | 114.30 | 8.250 | 209.55 |
| .063 | 1.59 | 2.125 | 53.98 | 4.562 | 115.87 | 8.375 | 212.73 |
| .094 | 2.39 | 2.250 | 57.15 | 4.625 | 117.48 | 8.500 | 215.90 |
| .125 | 3.18 | 2.281 | 57.94 | 4.750 | 120.65 | 9.000 | 228.60 |
| .187 | 4.75 | 2.375 | 60.33 | 4.844 | 123.04 | 9.250 | 234.95 |
| .188 | 4.78 | 2.500 | 63.50 | 4.875 | 123.83 | 9.375 | 238.13 |
| .213 | 5.41 | 2.625 | 66.68 | 4.969 | 126.21 | 9.500 | 241.30 |
| .281 | 7.14 | 2.719 | 69.06 | 5.000 | 127.00 | 9.625 | 244.48 |
| .391 | 9.93 | 2.750 | 69.85 | 5.094 | 129.39 | 10.125 | 257.18 |
| .438 | 11.13 | 2.875 | 73.03 | 5.125 | 130.18 | 10.375 | 263.53 |
| .625 | 15.88 | 3.000 | 76.20 | 5.250 | 133.35 | 10.750 | 273.05 |
| .812 | 20.62 | 3.062 | 77.77 | 5.375 | 136.53 | 11.000 | 279.40 |
| .844 | 21.44 | 3.125 | 79.38 | 5.500 | 139.70 | 11.250 | 285.75 |
| .906 | 23.01 | 3.188 | 80.98 | 5.750 | 146.05 | 12.625 | 320.68 |
| 1.062 | 26.97 | 3.250 | 82.55 | 5.875 | 149.23 | 12.875 | 327.03 |
| 1.125 | 28.58 | 3.375 | 85.73 | 6.000 | 152.40 | 13.125 | 333.38 |
| 1.188 | 30.18 | 3.500 | 88.90 | 6.250 | 158.75 | 13.500 | 342.90 |
| 1.250 | 31.75 | 3.625 | 92.08 | 6.375 | 161.93 | 13.750 | 349.25 |
| 1.344 | 34.14 | 3.656 | 92.86 | 6.500 | 165.10 | 15.625 | 396.88 |
| 1.375 | 34.93 | 3.750 | 95.25 | 6.750 | 171.45 | 16.750 | 425.45 |
| 1.469 | 37.31 | 3.812 | 96.82 | 6.875 | 174.63 | | |

NOTES:
1. Dimensions are in inches.
2. Shape of retainer optional provided specified dimensions are met. Spade lugs shall be rigid and double welded, double riveted, single riveted and soldered, or welded and single riveted to the retainer.
3. Minimum thickness of retainers shall be .045 (1.14 mm).
4. When two part numbers are applicable to the same retainer, both part numbers shall be marked on the retainer. Retainers shall be marked on the outer surface.
5. When used, rivets shall curl-over .020 (.51 mm) ±.015 (.38 mm) beyond the inside of the retainer.

FIGURE 2. CQ 072 retainers - Continued.



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NOTE: Style CQ72 capacitors, characteristics F and G, are substitutable for styles CP70 and CP72 capacitors, characteristics E and F, respectively, formerly covered in MIL-C-25/7E as shown in the following:

| FROM | TO | FROM | TO |
|---|---|---|---|
| MIL-C-25/7E (CP70 and CP72) | MIL-PRF-19978/3E (CQ72) | MIL-C-25/7E (CP70 and CP72) | MIL-PRF-19978/3E (CQ72) |
| CP7--1EF254K1 | CQ721EF254K1 | CP7-E1-K205K- | CQ72E1-K205K- |
| CP7--1EF105K1 | CQ721EF105K1 | CP7-E1EK405K- | CQ72E1-K405K- |
| CP7--1-F205K1 | CQ721-F205K1 | CP7-E1-K605K- | CQ72E1-K605K- |
| CP7--1-F405K1 | CQ721-F405K1 | CP7-E1-K805K- | CQ72E1-K805K- |
| CP7--1-F605K1 | CQ721-F605K1 | CP7-E1-K106K- | CQ72E1-K106K- |
| CP7--1-F805K1 | CQ721-F805K1 | CP70E1-K126K- | CQ72E1-K126K- |
| CP7--1-F106K1 | CQ721-F106K1 | CP7-E1-L104K1 | CQ72E1-L104K1 |
| CP7--1EG104K1 | CQ721EG104K1 | CP7-E1-L254K1 | CQ72E1-L254K1 |
| CP7--1EG254K1 | CQ721EG254K1 | CP7-E1-L504K1 | CQ72E1-L504K1 |
| CP7--1EG504K1 | CQ721EG504K1 | CP7-E1-L105K- | CQ72E1-L105K- |
| CP7--1-G105K1 | CQ721-G105K1 | CP7-E1-L205K- | CQ72E1-L205K- |
| CP7--1-G205K1 | CQ721-G205K1 | CP7-E1-L405K- | CQ72E1-L405K- |
| CP7--1-G405K1 | CQ721-G405K1 | CP7-E1-L605K- | CQ72E1-L605K- |
| CP7--1-G605K1 | CQ721-G605K1 | CP7-E1EM104K1 | CQ72E1FM104K1 |
| CP7--1-G805K1 | CQ721-G805K1 | CP7-E1-M254K- | CQ72E1-M254K- |
| CP7--1-G106K1 | CQ721-G106K1 | CP7-E1-M504K- | CQ72E1-M504K- |
| CP7--1-G126K1 | CQ721-G126K1 | CP7-E1-M105K- | CQ72E1-M105K- |
| CP7--1-G156K1 | CQ721-G156K1 | CP7-E1-M205K- | CQ72E1-M205K- |
| CP7--1EH104K1 | CQ721EH104K1 | CP7-E1-M405K- | CQ72E1-M405K- |
| CP7--1-H254K1 | CQ721-H254K1 | CP7-E1EN104K- | CQ72E1-N104K- |
| CP7--1EH504K1 | CQ721EH504K1 | CP7-E1EN254K- | CQ72E1-N254K- |
| CP7--1-H105K1 | CQ721-H105K1 | CP7-E1-N504K- | CQ72E1-N504K- |
| CP7--1-H205K1 | CQ721-H205K1 | CP7-E1-N105K- | CQ72E1-N105K- |
| CP7--1-H405K1 | CQ721-H405K1 | CP7-E1-N205K- | CQ72E1-N205K- |
| CP7--1-H605K1 | CQ721-H605K1 | CP70E1-N405K1 | CQ72E1-N405K1 |
| CP7--1-H805K1 | CQ721-H805K1 | CP7-D1-P104K- | CQ72D1-P104K- |
| CP7--1-H106K1 | CQ721-H106K1 | CP7-D1-P254K- | CQ72D1-P254K- |
| CP7--1-H126K1 | CQ721-H126K1 | CP7-D1-P504K- | CQ72D1-P504K- |
| CP7--1-H156K1 | CQ721-H156K1 | CP7-D1-P105K- | CQ72D1-P105K- |
| CP7--E1EJ254K1 | CQ72E1EJ254K1 | CP70D1-P205K- | CQ72D1-P205K- |
| CP7--E1EJ504K1 | CQ72E1EJ504K1 | CP7-D1-R104K- | CQ72D1-R104K- |
| CP7--E1-J105K1 | CQ72E1-J105K1 | CP7-D1-R254K- | CQ72D1-R254K- |
| CP7--E1-J205K1 | CQ72E1-J205K1 | CP7-D1-R504K- | CQ72D1-R504K- |
| CP7--E1-J405K1 | CQ72E1-J405K1 | CP70D1ER105K- | CQ72-1-R105K- |
| CP7--E1-J605K1 | CQ72E1-J605K1 | CP70D1-R205K- | CQ72-1-R205K- |
| CP7--E1-J805K1 | CQ72E1-J805K1 | CP70D1-S504K- | CQ72-1-S504K- |
| CP7--E1-J106K1 | CQ72E1-J106K1 | CP70D1-S105K- | CQ72-1-S105K- |
| CP7--E1-J126K1 | CQ72E1-J126K1 | CP70D1-S205K1 | CQ72-1-S205K1 |
| CP7--E1-J156K1 | CQ72E1-J156K1 | CP70D1ET104K- | CQ72-1-T104K- |
| CP7--E1EK104K1 | CQ72E1EK104K1 | CP70D1ET254K- | CQ72-1-T254K- |
| CP7--E1-K254K1 | CQ72E1-K254K1 | CP70D1-T504K- | CQ72-1-T504K- |
| CP7--E1-K504K1 | CQ72E1-K504K1 | CP70D1-T105K- | CQ72-1-T105K- |
| CP7--E1-K105K1 | CQ72E1-K105K1 | CP70D1-T205K1 | CQ72-1-T205K1 |

Custodians:
Army - CR
Navy - EC
Air Force - 11
DLA - CC

Review activities:
Navy - MC
Air Force - 19, 99

Preparing activity:
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